RANGE 42

Open-source cyber range platform for collaborative security training

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RANGE 42 Team :

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WHAT IS RANGE 42 ?

RANGE 42 modular cyber range platform for **building** and **deploying** realistic cyber training infrastructures on-premises.

This project developed in collaboration with **NC3**

The project is still at an early stage, but it's open source and contributions are welcome!

FEATURES OVERVIEW

Currently **RANGE42** provide two main capabilities :

- **Deploy** vulnerable and misconfigured hosts
- **Include an extensible catalogue** of ready-to-deploy CVEs, misconfigured services, and product setups

Coming Soon :

- **Web Infrastructure designer UI :** draw a network and deploy it automatically
- **Expand the catalogue** from 20 to over 50 to ready to deploy scenarios

FOR WHO ? WHY IT'S USEFUL ?

Our goal is to let anyone in cybersecurity test, break, and analyse realistic scenarios

- from system hardening to full compromise and incident response.

A few examples:

- **Sysadmins / Network admins** could practice securing vulnerable stacks and test hardening procedures
- **SOC analysts / Blue teams** could validate detection rules, tune alerts and test Incident response workflows
- **Red teamers / Security researcher**s could build exploit chains or study CVEs in controlled environments
- **Forensics teams / investigators** could reconstruct incidents and analyse compromised systems through practical drills

ARCHITECTURE OVERVIEW

Hypervisor layer :

- Proxmox

Automation layer :

- Ansible to handles provisioning, configuration and orchestration of systems

Container layer :

- Docker / LXC to run specific services, misconfigurations or application stacks

Monitoring & telemetry layer :

- Wazuh to collects events and detection data

Network & Access & isolation layer :

- Firewalls, VPN layers to ensure secure and segmented lab access

Web applications & API layer :

- VueJS, Python3, Fast API with Kong as API gateway

HOW TO CONTRIBUTE ?

The project still is in an early stage, the main contributions paths are :

Scenario contributors :

- **CVE researchers**
- **Misconfig labs**
- **Documentation**
- **...**

Infrastructure automation :

- **Ansible roles / playbooks**
- **CI/CD pipelines**
- **Backend API features:**
- **...**

UI / UX design :

- **Lab designer**
- **User workflow**
- **...**

You are devops, pentester, UI designer, ...
feel free to contribute !

WHERE TO FIND THE REPOs ?

https://www.range42.lu